

7	8	6	1	2
0	1	2	3	4

A for loop will traverse of unsorted/sorted array.

During traversal $arr[i] == \text{key}$ then return key value.

Even, finding a number which are not in array. then it will return -1;

```

Transaction.java
1  import java.util.Scanner;
2  public class Transaction{
3
4  public static int linearSearch( int TransactionId[] , int SuspiciousId ){
5
6      for(int i=0; i<TransactionId.length; i++){
7          if(TransactionId[i]==SuspiciousId){
8              return i;
9          }
10     }
11     return -1;
12 }
13 public static void main(String[] args){
14     Scanner sc = new Scanner(System.in);
15
16     System.out.print("Enter number of Transaction: ");
17     int n =sc.nextInt();
18
19     System.out.println("Enter Transaction Ids(Space seperated): ");
20     int TransactionId[] = new int[n];
21
22     for(int i =0; i<n; i++){
23         TransactionId[i] = sc.nextInt();
24     }
25     System.out.print("Enter Suspicious Ids: ");
26     int SuspiciousId = sc.nextInt();
27     int index =linearSearch(TransactionId , SuspiciousId);
28
29     if(index!=-1){
30         System.out.println("Fraud Transaction Found");
31     }
32     else{
33         System.out.println("Transaction Not Found");
34     }
35 }
36 }

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

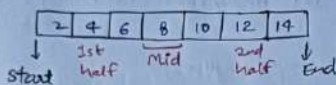
```

PS C:\Users\Aashish\OneDrive\Desktop\java> cd "c:\Users\Aashish\OneDrive\Desktop\jav
Enter number of Transaction: 5
Enter Transaction Ids(Space seperated):
1201 1305 1450 1408 1502
Enter Suspicious Ids: 1450
Fraud Transaction Found
PS C:\Users\Aashish\OneDrive\Desktop\java>

```

Binary Search

Prerequisite → Sorted Arrays



if (Key < mid) → check on first half
if (Key > mid) → check on 2nd half

for first half :-



2nd half :-



Pseudo code -

start = 0, end = n-1

while (start <= end)

find mid → (start+end)/2

Compare mid & Key

mid == Key Found

mid > Key Left

mid < Key Right

```
Serverlog.java
1 import java.util.Scanner;
2 public class Serverlog {
3
4     public static int binarysearch(int logIds[], int targetlogids){
5
6         int start = 0; int end = logIds.length-1;
7         while(start<=end)
8         {
9             int mid =(start+end)/2;
10            if(logIds[mid]==targetlogids){
11                return mid;
12            }
13            if(logIds[mid] < targetlogids){
14                start=mid+1;
15            }
16            else{
17                end =mid-1;
18            }
19        }
20        return -1;
21    }
22
23    public static void main(String[] args){
24
25        Scanner sc = new Scanner(System.in);
26        int n = sc.nextInt();
27
28        int logIds[] = new int[n];
29
30        for(int i = 0; i<n; i++){
31            logIds[i] = sc.nextInt();
32        }
33
34        int targetlogids = sc.nextInt();
35        binarysearch(logIds,targetlogids);
36        System.out.println("Log Found" );
37    }
38 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\Aashish\OneDrive\Desktop\java> cd "c:\Users\Aashish\OneDrive\Desktop\

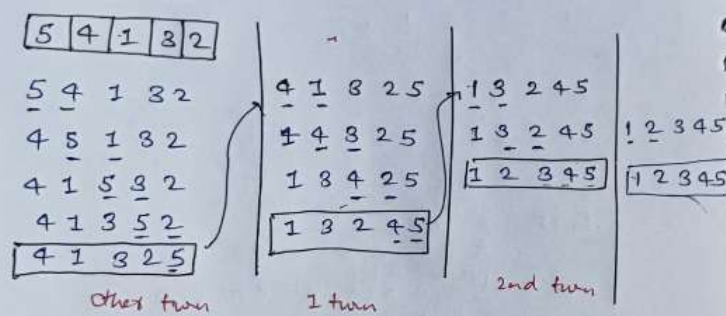
7

1001 1005 1010 1015 1020 1030 1040

1015

Log Found

PS C:\Users\Aashish\OneDrive\Desktop\java>



Time Complexity $\rightarrow O(n^2)$

```

Employee.java
1  import java.util.Scanner;
2  public class Employee{
3
4      public static void bubblesort(int salaries[] ){
5
6          for(int i =0; i < salaries.length-1; i++) {
7              for(int j = 0; j<salaries.length-1-i; j++)
8              {
9                  if(salaries[j] > salaries[j+1]){
10                     int temp = salaries[j];
11                     salaries[j] = salaries[j+1];
12                     salaries[j+1] = temp;
13                 }
14             }
15         }
16     }
17     public static void printsalaries(int salaries[]){
18         for(int i =0; i<salaries.length; i++){
19             System.out.print(salaries[i]+" ");
20         }
21         System.out.println();
22     }
23     public static void main(String args[]){
24         Scanner sc = new Scanner(System.in);
25
26         int n = sc.nextInt();
27         int salaries[] = new int[n];
28
29         for(int i = 0; i<n; i++){
30             salaries[i] = sc.nextInt();
31         }
32         bubblesort(salaries);
33         System.out.print("Sort Employee salaries: ");
34         printsalaries(salaries);
35     }
36 }

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```

PS C:\Users\Aashish\OneDrive\Desktop\java> cd "c:\Users\Aashish\OneDrive\Desktop
5
45000 32000 45000 55000 60000
Sort Employee salaries: 32000 45000 45000 55000 60000
PS C:\Users\Aashish\OneDrive\Desktop\java>

```

Insertion Sort:-

Time Complexity $\rightarrow O(n^2)$

Idea $\rightarrow$ Pick an element (from unsorted part) & place in the right pos in sorted part.

5 4 1 3 2

int curr = arr[i];
int prev = i-1;

5	4	1	3	2
0	1	2	3	4

insertion sorting me 0 index ko sorted part man lete.

arr[i] = 4 ko hum apni curr (value)
prev = arr[0] = 5

while(prev $\geq$ 0 && arr[prev] > arr[i])

arr[0] > arr[1] / 5 > 4

arr[prev+1] = arr[prev] prev--;

insert

arr[prev+1] = curr;

When prev = -1
loop ruk jayega.

prev = -1 + 1
= 0

arr[0] = curr;

4, 5, 1, 3, 2

```
1 import java.util.Scanner;
2 public class Websiteresponse
3 {
4     public static void insertionsort (int response[]){
5
6         for(int i = 1; i < response.length; i++){
7             {
8                 int curr = response[i];
9                 int prev = i-1;
10                while(prev >= 0 && response[prev] > curr){
11                    response[prev+1] = response[prev];
12                    prev--;
13                }
14                response[prev+1] = curr;
15            }
16        }
17
18        public static void printresponse(int response[])
19        {
20            for(int i=0; i<response.length; i++){
21                System.out.print(response[i]+" ");
22            }
23            System.out.println();
24        }
25
26        public static void main(String args[]){
27
28            Scanner sc = new Scanner(System.in);
29            int n = sc.nextInt();
30
31            int response[] = new int[n];
32            for(int i = 0; i<n; i++){
33                response[i] = sc.nextInt();
34            }
35            insertionsort(response);
36            System.out.print("Sorted response times : ");
37            printresponse(response);
38        }
39    }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\Aashish\OneDrive\Desktop\java> cd "c:\Users\Aashish\OneDrive\Desktop\j
5
250 120 320 180 100
Sorted response times : 100 120 180 250 320
PS C:\Users\Aashish\OneDrive\Desktop\java>
```

Selection sort:-

5 4 1 2 3

5 4 1 2 3

1 4 5 2 3

1 2 5 4 3

1 2 3 4 5

1 2 3 4 5

Time Complexity $\rightarrow O(n^2)$

logic -

ek outer loop hoga.

ek inner loop jisme hum

i ke 0 index se j ki

value tak loop chlega.

Jo chhota hoga vhi

hamara minPos hoga.

ab inner loop se bahar

nikaile current value [i]

and minPos ko swap kra

denge.

Ex:-

2 6 3 7 8 5

2 6 3 7 8 5

2 3 6 7 8 5 2 3 6 7 8 5

2 3 6 7 8 5 2 3 5 6 7 8

2 3 6 7 8 5 2 3 5 6 7 8

2 3 5 6 7 8

Priorityscores.java

```
1 import java.util.Scanner;
2 public class Priorityscores {
3
4     public static void selectionsort(int priorityscores[]){
5         for(int i=0; i<priorityscores.length-1; i++){
6             int minpos = i;
7             for(int j=i+1; j<priorityscores.length; j++)
8             {
9                 if(priorityscores[minpos] > priorityscores[j])
10                {
11                    minpos = j;
12                }
13            }
14            int temp = priorityscores[minpos];
15            priorityscores[minpos] = priorityscores[i];
16            priorityscores[i] = temp;
17        }
18    }
19    public static void printpriorityscores (int priorityscores[]){
20        for(int i=0; i<priorityscores.length; i++){
21            System.out.print(priorityscores[i]+" ");
22        }
23        System.out.println();
24    }
25    public static void main (String args[]){
26        Scanner sc = new Scanner(System.in);
27        int n = sc.nextInt();
28        int priorityscores[] = new int[n];
29
30        for(int i = 0; i<n; i++){
31            priorityscores[i] = sc.nextInt();
32        }
33        selectionsort(priorityscores);
34        System.out.print("Sorted priorityscore : ");
35        printpriorityscores(priorityscores);
36    }
37 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
P5 C:\Users\Aashish\OneDrive\Desktop\java> cd "c:\Users\Aashish\OneDrive\Desktop\java\" ; if
5
9 2 8 1 5
Sorted priorityscore : 1 2 5 8 9
P5 C:\Users\Aashish\OneDrive\Desktop\java>
```